

Sabyasachi Chatterjee

ASSOCIATE PROFESSOR OF STATISTICS

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Research Interests

Nonparametric Statistics, Machine Learning, High Dimensional Statistics, Applied Probability, Sequential Decision Making, Statistical Information Theory.

Education

Yale University

PHD STATISTICS

- Advisor: Dr. Andrew Barron

New Haven, USA

2009-2014

Indian Statistical Institute

MASTERS IN STATISTICS

Kolkata, India

2007 - 2009

Indian Statistical Institute

BACHELOR'S IN STATISTICS

Kolkata, India

2004 - 2007

Professional Experience

- 2024-2026 **Associate Professor**, University of Illinois at Urbana Champaign
- 2017-2024 **Assistant Professor**, University of Illinois at Urbana Champaign
- 2015-2017 **William Kruskal Instructor**, Department of Statistics, University of Chicago, Advised by Dr. John Lafferty.
- 2014-2015 **Postdoctoral Research Scholar**, Department of Statistics, University of Chicago, Advised by Dr. John Lafferty.
- 2011-2013 **Graduate Teaching Assistant**, Department of Statistics and Data Science, Yale University

Publications

PUBLISHED

- C.M Madrid-Padilla, O.H. Madrid-Padilla, **S. Chatterjee**. Risk Bounds For Distributional Regression. To appear in NeurIPS 2025.
- D. Bhattacharyya, E. Thiemann, J. Machol, G.Cucho-Padin, **S. Chatterjee**, W. Harris, E. Mierkiewicz. The hydrogen Lyman α line shape in the exospheres of terrestrial objects in the solar system. Frontiers in Astronomy and Space Sciences, 2025. Volume 12.
- S. Chatterjee**, P-S. Dey and S. Goswami. Central Limit Theorem for Gram-Schmidt Random Walk Design. Annals of Applied Probability, 35(2): 1407-1441 (April 2025).
- O.-H. M-Padilla and **S. Chatterjee**. Quantile Regression by Dyadic CART. Electronic Journal of Statistics, 8(1), 1206-1247, 2024.
- A. Chaudhuri and **S. Chatterjee**, A Cross Validation Framework for Signal Denoising with Applications to Trend Filtering, Dyadic CART and Beyond. Annals of Statistics Vol. 51, No. 4. DOI: 10.1214/23-AOS2283.
- S. Chatterjee** and S. Goswami, Spatially Adaptive Online Prediction of Piecewise Regular Functions. Algorithmic Learning Theory 2023, 35 pp., Proc. Mach. Learn. Res. (PMLR), 201, 2023.
- T. Zhang and **S. Chatterjee**, Element-wise Estimation Error of Generalized Fused Lasso. Bernoulli, 29(4), 2691-2718, 2023.

- Y. Yu, **S. Chatterjee** and H. Xu, Localizing Change Points in Piecewise Polynomials of General Degrees, *Electronic Journal of Statistics*, Vol. 16, p. 1855 – 1890, 2022.
- O. H. M. Padilla and **S. Chatterjee**, Risk Bounds for Quantile Trend Filtering, *Biometrika*, 2021.
- S. Chatterjee** and S. Goswami, Adaptive Estimation of Multivariate Piecewise Polynomials and Bounded Variation Functions by Optimal Decision Trees, *Annals of Statistics*, Vol. 49, p. 2531 – 2551, 2021.
- S. Chatterjee** and S. Goswami, New Risk Bounds in 2D Total Variation Denoising, *IEEE Transactions of Information Theory*, Vol. 67, p. 4060 – 4091, 2021.
- A. Guntuboyina, D. Lieu, **S. Chatterjee** and B. Sen, Adaptive Risk Bounds in Univariate Total Variation Denoising and Trend Filtering, *Annals of Statistics*, Vol. 48, p. 205 – 229, 2020.
- S. Chatterjee** and S. Mukherjee, Estimation in Tournaments and Graphs Under Monotonicity Constraints, *IEEE Transactions on Information Theory*, Vol. 65, Issue: 6, doi:10.1109/TIT.2019.2893911, 2019.
- Q. Han, T. Wang, **S. Chatterjee** and R. Samworth, Isotonic Regression in General Dimensions, *Annals of Statistics*, Vol. 43, Number 5, p. 2440 – 2471, 2019.
- S. Chatterjee** and J. Lafferty, Adaptive Risk Bounds in Unimodal Regression, *Bernoulli*, Vol. 25, Number 1, p. 1 – 25, 2019.
- M. Bonakdarpour, **S. Chatterjee**, R. Barber, J. Lafferty, Prediction Rule Reshaping, 35th International Conference on Machine Learning (ICML 2018), 2018.
- S. Chatterjee** and J. Lafferty, Denoising Flows on Trees, *IEEE Transactions on Information Theory*, Vol. 64, Issue: 3, 2018.
- S. Chatterjee**, A. Guntuboyina and B. Sen, On Matrix Estimation Under Monotonicity Constraints, *Bernoulli*, Vol. 24, p. 1072 – 1100, 2018.
- Y. Zhu, **S. Chatterjee**, J. Duchi and J. Lafferty, Local Minimax Complexity of Stochastic Convex Optimization, *Neural Information Processing Systems (NeurIPS 2016)*, p. 3423 – 3431, 2016.
- S. Chatterjee**, An Improved Global Risk Bound in Concave Regression, *Electronic Journal of Statistics*, 10.1, p. 1608 – 1629, 2016.
- S. Chatterjee**, A. Guntuboyina, and B. Sen, On Risk Bounds in Isotonic and Other Shape Restricted Regression Problems, *Annals of Statistics*, Vol. 43, p. 1774 – 1800, 2015.
- S. Chatterjee** and A. Barron, Information Theoretic Validity of Penalized Likelihood, 2014 IEEE International Symposium on Information Theory, ISIT 2014, p. 3027 – 3031, 2014.

IN REVIEW

- S. Chatterjee** and S. Sen, Regret Minimization in Isotonic, Heavy Tailed Contextual Bandits via Adaptive Confidence Bands. Available at <https://arxiv.org/pdf/2110.10245.pdf>.
- S. Chatterjee**, S. Goswami and S.S. Mukherjee, LASER: A new approach to locally adaptive polynomial regression. Available at <https://arxiv.org/abs/2412.19802>.
- S. Chatterjee**, Minmax Trend Filtering: A Locally Adaptive Nonparametric Regression Method via Pointwise Min Max Optimization. Available at <https://arxiv.org/abs/2410.03041>.
- D. Ghoshal and **S. Chatterjee**, An Exact Pointwise Characterization for Total Variation Denoising in Quantile Regression. Available at <https://arxiv.org/abs/2605.01237>.

Grants, Honors & Awards

2019-2023	Nonparametric Estimation Under Shape/Norm Constraints , National Science Foundation Department of Mathematical Sciences 1916375	\$ 160,000
Fall 2022	Rated Excellent Teacher , University of Illinois at Urbana Champaign	
Spring 2021	Rated Excellent Teacher , University of Illinois at Urbana Champaign	
Spring 2020	Rated Excellent Teacher , University of Illinois at Urbana Champaign	
2009	Sterling Fellowship Award , Yale University	
2007	National Board of Higher Mathematics Award , Government of India	

Presentations

INVITED TALKS

- "A New Minmax Representation for Total Variation Denoising", Marshall School of Business, University of Southern California, April 2026.
- "A New Minmax Representation for Total Variation Denoising", Department of Applied Mathematics and Statistics, Colorado Mines University, April 2026.
- "Revisiting Total Variation Denoising: New Formula, Generalizations and Insights", International Indian Statistical Association, Lincoln, Nebraska, June 2025.
- "Revisiting Total Variation Denoising: New Formula, Generalizations and Insights", International Indian Statistical Association, Lincoln, Nebraska, June 2025.
- "Revisiting Total Variation Denoising: New Formula, Generalizations and Insights", Cornell University, Department of Statistics and Data Science, April 2025.
- "LASER: A New Locally Adaptive Nonparametric Regression Method", International Indian Statistical Association, Kochi, India, December 2024.
- "A New Locally Adaptive Nonparametric Regression Method", Indian Institute of Science, Department of Computational Data Sciences, Bengaluru, August 2024.
- "A New Locally Adaptive Nonparametric Regression Method", Google Research India, Bengaluru, August 2024.
- "A New Locally Adaptive Nonparametric Regression Method", Indian Statistical Institute, Statistics and Mathematics Unit, Bengaluru, August 2024.
- "Theory for K Fold Cross Validation in Nonparametric Estimation", Washington University at St. Louis, Department of Statistics, January 2024.
- "Theory for K Fold Cross Validation in Nonparametric Estimation", Indian Statistical Institute, Interdisciplinary Statistical Research Unit, Kolkata, January 2024.
- "Theory for K Fold Cross Validation in Nonparametric Regression", University of Chicago, Department of Statistics, October 2023.
- "Theory for K Fold Cross Validation in Nonparametric Regression", University of California at Davis, Department of Statistics, October 2023.
- "Theory for K Fold Cross Validation in Nonparametric Regression", Georgia Institute of Technology, School of Industrial Systems and Engineering, August 2023.
- "A Theoretically Tractable Framework for K Fold Cross Validation", University of Pennsylvania, Wharton School of Statistics, April 2023.
- "A Theoretically Tractable Framework for K Fold Cross Validation", University of California at Los Angeles, April 2023.
- "Locally Adaptive Online Prediction of Piecewise Polynomial Signals", Algorithmic Learning Theory, Singapore, February 2023.
- "A Theoretically Tractable Framework for K Fold Cross Validation" Google Research, India, February 2023.
- "A Theoretically Tractable Framework for K Fold Cross Validation", Tata Institute of Fundamental Research, Computer Science Department, Mumbai, February 2023.
- "A Theoretically Tractable Framework for K Fold Cross Validation", Indian Institute of Technology, Mumbai, February 2023.
- "A Theoretically Tractable Framework for K Fold Cross Validation", Indian Statistical Institute, January 2023.
- "Locally Adaptive Online Prediction of Piecewise Polynomial Signals via Sleeping Experts Aggregation over Rectangles", International Indian Statistical Association, Bengaluru, December 2022.
- "Pointwise Bounds for Fused Lasso", CM Statistics, Virtual Meeting, December 2022.
- "A Theoretically Tractable Framework for K Fold Cross Validation", Ecostat 2022, Virtual Meeting, June 2022.
- "A Theoretically Tractable Framework for K Fold Cross Validation", International Centre for Mathematical Sciences Workshop on Structural Breaks and Shape Constraints, Virtual Seminar, May 2022.

“A Theoretically Tractable Framework for K Fold Cross Validation”, UIUC Purdue Joint Seminar, April 2022.

“A Theoretically Tractable Framework for K Fold Cross Validation”, University of Washington, St. Louis, Mathematics Department, Virtual Seminar, February 2022.

“Dyadic CART: An alternative to Trend Filtering”, Joint Statistical Meetings, Virtual meeting, August 2021.

“Adaptive estimation of piecewise rectangular signals”, International Indian Statistical Association, Virtual Meeting, May 2021.

“Dyadic CART: An alternative to Trend Filtering”, CM Statistics, Virtual meeting, December 2020.

“Adaptive estimation of multivariate piecewise polynomials by optimal decision trees”, Joint Statistical Meetings, Virtual meeting, August 2020.

“Adaptive estimation by optimal decision trees”, International Indian Statistical Association, Mumbai, India, December 2019.

“Adaptive estimation of piecewise constant and bounded variation matrices by optimal decision trees”, CM Statistics, London, UK, December 2019

“Adaptive estimation of multivariate piecewise constant functions by Dyadic CART type methods”, Ecostat 2019, Taichung, Taiwan, June 2019

“Adaptive risk bounds in the nonparametric Bradley Terry model”, Business School in University of Miami, Miami, Florida, September 2018

“Adaptive risk bounds for 2D total variation denoising”, Statslab, Cambridge University, UK, June 2018

“Risk bounds for 2D total variation denoising”, International Indian Statistical Association, Gainesville, Florida, May 2018

“Risk bounds for 2D total variation denoising”, Newton Institute, UK, April 2018

“Isotonic regression in general dimensions”, Workshop on Shape Constrained Methods in Statistics, Banff, Canada, January 2018

“Estimation in tournaments and graphs under monotonicity constraints”, Workshop on Networks, Indian Statistical Institute, Kolkata, India, January 2018

“Isotonic regression in general dimensions”, Joint Statistical Meetings, Baltimore, Maryland, August 2017

“Adaptation in trend filtering”, IMS China, Nanning, China, June 2017

“Estimation of pairwise probabilities in tournaments under monotonicity constraints”, Statistics Department at University of Iowa, Iowa, September 2016

“Denoising flows on trees”, International Indian Statistical Association, Corvallis, Oregon, August 2016

“Estimation of pairwise probabilities in tournaments under monotonicity constraints”, Workshop on Shape Constrained Methods, Oberwolfach, Germany, July 2016

“Adaptive risk bounds in unimodal regression”, International Society of Nonparametric Statistics, Avignon, France, June 2016

Teaching Experience _____

Spring 2025 **Advanced Topics in Statistical Learning**, Graduate Level *UIUC*

Spring 2025 **Actuarial Statistics**, Undergraduate Level *UIUC*

Fall 2024 **Advanced Mathematical Statistics**, Graduate Level *UIUC*

Fall 2023 **Introduction to High Dimensional Probability with Applications to Data Science**, Graduate Level *UIUC*

Fall 2023 **Introduction to Empirical Process Theory with Applications**, Graduate Level *UIUC*

Fall 2022 **Introduction to Stochastic Processes**, Undergraduate Level *UIUC*

Fall 2022 **Introduction to Online Learning Theory**, Graduate Level *UIUC*

Spring 2022 **High Dimensional Probability**, Graduate Level *UIUC*

Fall 2021 **Introduction to Stochastic Processes**, Undergraduate Level *UIUC*

Spring 2021 **Empirical Process Theory and Weak Convergence**, Graduate Level *UIUC*

Fall 2020 **Introduction to Statistics and Probability 2**, Undergraduate Level *UIUC*

Spring 2020 **Large Sample Theory**, Graduate Level *UIUC*

Fall 2019 **Introduction to Statistics and Probability 1**, Undergraduate Level *UIUC*

Spring 2019 **Modern Statistical Estimation Theory**, Graduate Level *UIUC*

Fall 2018 **Introduction to Stochastic Processes**, Undergraduate Level *UIUC*

Spring 2018 **Mathematical Statistics**, Graduate Level *UIUC*

Fall 2017 **Introduction to Statistics and Probability 1**, Undergraduate Level *UIUC*

Advising _____

Ph.D. Dissertation Advisor of

1. Deep Ghoshal (5th year Phd Student)

Masters Thesis Advisor of

1. Arka Sinha (Masters student in Indian Statistical Institute).

Outreach & Professional Development _____

SERVICE AND OUTREACH

2020-2023	Blackwell Scholarship Program , Research Mentor	<i>UIUC</i>
2021	IMS Invited Session, “Trend Filtering and Related Nonparametric Regression Methods”, Joint Statistical Meetings , Organizer	<i>Virtual Meeting</i>
2021	IMS Invited Session, “Recent Advances in Trend Filtering and Related Methods”, International Indian Statistical Association , Organizer	<i>Virtual Meeting</i>
2022	IMS Invited Session, “Recent Advances in Online Methods, International Indian Statistical Association , Organizer	<i>Bangalore, India</i>

EDITORSHIP

I am currently an Associate Editor of Sankhya A.

PEER REVIEW

Annals of Statistics
 IEEE Transactions of Information Theory
 Bernoulli
 Journal of American Statistical Association
 Electronic Journal of Statistics
 SIAM Journal of Mathematics of Data Science
 Biometrika
 Information and Inference
 Mathematical Statistics and Learning
 Journal of the Royal Statistical Society: Series B.

PROFESSIONAL MEMBERSHIPS

American Statistical Association
 International Indian Statistical Association